CS4433 – Moderated Learning Management System

System and Acceptance Testing

1. System Testing

1.1 Purpose

System testing verifies that the complete Moderated Learning Management System operates correctly as an integrated application.

Testing is performed using the compiled application and its SQLite database. The application is accessed through the console interface using the four supported user roles:

- Student
- Lecturer
- Head of Department
- Administrator

The tests cover authentication, role-specific functionality, course management, student enrolment, assignment management, submissions, grading, moderation and result release.

1.2 System Test Cases

The following test cases are derived from the main LMS functionality and user workflows.

Test ID	Test scenario	Expected result	Actual result / status
ST-01	Log in using valid Student credentials.	Student is authenticated and the Student menu is displayed.	Successful login demonstrated. **Passed**
ST-02	Log in using valid Lecturer credentials.	Lecturer is authenticated and the Lecturer menu is displayed.	Successful login demonstrated. **Passed**
ST-03	Log in using valid Administrator credentials.	Administrator is authenticated and the Administrator menu is displayed.	Successful login demonstrated. **Passed**
ST-04	Log in using incorrect credentials.	Login is rejected and the user returns to the login menu.	Rejection demonstrated. **Passed**
ST-05	Enter an invalid menu option.	Application displays an error and requests another choice.	Invalid option was rejected. **Passed**
ST-06	Enter alphabetic characters where a numeric menu choice is required.	Application displays an invalid input message without crashing.	Invalid input handling demonstrated. **Passed**

ST-07	Administrator creates a Student account.	Account is created and stored in the database.	Account creation demonstrated. **Passed**
ST-08	Newly created Student logs in.	Student can authenticate using the newly created credentials.	Successful login demonstrated. **Passed**
ST-09	Student views available courses.	Available courses are displayed with their course information.	Courses displayed. **Passed**
ST-10	Student enrolls in an available course.	Enrolment is recorded and the course is no longer available for enrolment by that student.	Three course enrolments demonstrated. **Passed**
ST-11	Student views enrolled-course assignments.	Assignments belonging to enrolled courses are displayed.	Assignments displayed. **Passed**
ST-12	Student submits an assignment using a valid file path.	Submission is stored and receives Submitted status.	Successful submission demonstrated. **Passed**
ST-13	Student submits an invalid or unavailable assignment ID.	Application rejects the assignment and displays an appropriate message.	Invalid assignment ID rejection demonstrated. **Passed**
ST-14	Student views an assignment with an approved individual extension.	Revised deadline is displayed for that student.	Revised deadline displayed as 15 November 2026. **Passed**
ST-15	Student without an extension views the same assignment.	Original deadline remains displayed.	Original deadline displayed as 10 November 2026. **Passed**
ST-16	Lecturer creates an assignment.	Assignment is saved and becomes available to eligible students.	Not recorded as executed. **Pending**
ST-17	Lecturer grants an individual extension.	Revised deadline and reason are recorded for the selected student.	Extension functionality previously demonstrated. **Passed**
ST-18	Lecturer grades a student submission.	Mark and feedback are recorded and submission status changes to Graded.	Grading previously demonstrated. **Passed**
ST-19	Head of Department moderates a graded submission.	Moderated mark is recorded and status changes to Moderated.	Moderation previously demonstrated. **Passed**
ST-20	Head of Department releases a moderated result.	Submission status changes to Released.	Result release previously demonstrated. **Passed**
ST-21	Student views results before release.	Unreleased marks are not displayed as released results.	Visibility behavior demonstrated for a submission. **Passed**

ST-22	Student views results after release.	Released mark and feedback are displayed.	Released result displayed successfully. **Passed**
ST-23	Student attempts to submit using an invalid file path.	Application rejects the invalid path and does not create an invalid submission.	Not recorded as executed. **Pending**
ST-24	Student attempts to view another student's submissions.	Access is restricted to the authenticated student's own records.	Not recorded as executed. **Pending**
ST-25	Database is unavailable when the application starts.	Application reports the database error safely.	Not recorded as executed. **Pending**
ST-26	Verify that a mark change creates an audit record.	Audit log records the old mark, new mark, user, time and reason.	Not recorded as executed. **Pending**

1.3 System Testing Results

The demonstrated system tests confirm that the main authentication, account creation, course enrolment, assignment viewing and submission functions operate successfully.

The individual extension test also confirms that assignment deadlines are retrieved according to the student's extension record.

The grading, moderation and release workflow was previously demonstrated using a late submission. The submission progressed through grading, moderation and release, and the student was subsequently able to view the released result.

Some system tests remain pending, particularly invalid file-path handling, cross-student access restrictions, database failure handling and audit-log verification.

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2. Acceptance Testing

2.1 Purpose

Acceptance testing evaluates whether the LMS satisfies the practical needs of its intended users.

The tests are based on the main user activities and expected outcomes of the system.

The acceptance scenarios represent the four main user roles and evaluate whether each role can complete its expected activities.

2.2 Acceptance Test Cases

Test ID	User role	Acceptance scenario	Expected result	Status
AT-01	Student	Log in, view available courses and enroll in a course.	Student can access the system, view available courses and successfully enroll.	**Passed**
AT-02	Student	View assignments and submit an assignment.	Student can view assignments for enrolled courses and submit a file successfully.	**Passed**
AT-03	Lecturer	Log in and manage assignment activities.	Lecturer can access lecturer functionality, create assignments, view submissions and grade them.	Partially demonstrated; further verification required
AT-04	Lecturer	Grant an individual extension to a student.	Lecturer can grant an extension and the affected student sees the revised deadline.	**Passed**
AT-05	Head of Department	Review, moderate and release student marks.	HOD can review graded submissions, moderate marks and release results.	**Passed**
AT-06	Administrator	Create and manage user accounts.	Administrator can create user accounts and the new users can log in.	**Passed**
AT-07	Student	View released academic results.	Student can access marks and feedback after results have been released.	**Passed**
AT-08	Student	Attempt to access another student's academic records.	System prevents access to another student's private records.	Pending verification
AT-09	System	Apply the late submission penalty policy.	Late submissions receive the correct penalty based on the number of days late.	Previously demonstrated for a submission more than three days late; boundary cases pending

2.3 Acceptance Testing Evaluation

The demonstrated acceptance scenarios show that the LMS supports important academic activities.

Students can log in, view courses, enroll in courses, access assignments and submit work. Lecturers can grant individual extensions, and the revised deadline is correctly reflected for the affected student.

The moderation workflow supports review and release of marks by the Head of Department. Students can subsequently view released marks and feedback.

Administrators can create Student accounts, and newly created accounts can be used to log in successfully.

However, acceptance of the entire system should remain conditional on completing the outstanding verification scenarios, especially record-access restrictions, the complete lecturer workflow and late-penalty boundary cases.

3. Defect and Issue Log

The following issues were identified during testing or require further investigation.

Defect ID	Description	Severity	Current status
D-01	Resubmission behavior for the same student and assignment requires clarification and verification.	High	Open
D-02	Assignment 2 displays an invalid due-date value of y.	Medium	Open
D-03	Additional branch testing is needed to improve the measured branch coverage of 40.7%.	Low	Improvement required
D-04	Invalid file-path handling has not yet been verified through system testing.	Medium	Pending verification
D-05	Cross-student submission access restrictions require explicit system testing.	High	Pending verification

3.1 Defect Retesting

When a defect is corrected, the related test case should be repeated.

The retest should confirm that:

1. The original problem no longer occurs.
2. The expected behavior is achieved.
3. The correction has not introduced new defects.
4. Related functionality continues to operate correctly.

The extension deadline display issue was corrected and retested. The student with an approved extension saw the revised deadline, while a student without an extension saw the original deadline.

4. Overall System and Acceptance Testing Conclusion

System and acceptance testing demonstrate that the Moderated LMS supports several essential academic workflows.

The application successfully demonstrated authentication for different user roles, Student account creation, course enrolment, assignment viewing, assignment submission and individual extension handling.

The grading, moderation and result-release workflow was also demonstrated, including the ability of students to view released marks and feedback.

The results provide evidence that the main application workflows are functioning. However, some planned test cases remain outstanding, and the identified issues must be considered before declaring the entire system fully verified and accepted.

The system is therefore considered to have demonstrated its core functionality, subject to the remaining verification activities.